



Causal dimension reduction for multidimensional continuous exposures

an application to environmental mixtures analysis

Thomas Hsiao, Ph.D.

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Emory University

Rollins School of Public Health

Department of Biostatistics and Bioinformatics

Introduction

Goals of Mixtures Analysis

- **Understand combined effects**

Assess how multiple correlated exposures jointly impact health.

- **Identify important contributors**

Distinguish exposures (or groups) driving associations.

- **Handle correlation and interactions**

Account for complex dependencies among exposures.

- **Improve causal interpretation**

Define meaningful targets (e.g., intervention on the mixture).

- **Support policy and intervention**

Inform regulations by evaluating effects of realistic multi-exposure scenarios.

- Weighted quantile sum regression (WQS)
- Quantile g -computation (q -gcomp)
- Bayesian kernel machine regression (BKMR)
- Bayesian additive regression trees (BART)
- Distributed lag models with regression trees (DLM tree)
- Bayesian multiple index models (BMIM)

Limitations

Out of all the previous methods, only q -gcomp is causal, but it discards information and makes unrealistic parametric assumptions.

The other full exposure-response methods are not causal and can be hard to interpret.

Proposed solution: causally sufficient PCA (csPCA)

We propose a method that can be both **causal** and **interpretable** without throwing away too much information.

The causal inference literature tells us how to account for confounding while making minimal assumptions.

The sufficient dimension reduction (SDR) literature tells us how to do supervised dimension reduction.

Causal inference

- Observed data: $\mathbf{O} = (Y, \mathbf{X}, \mathbf{C}) \in \mathcal{Y} \times \mathcal{X} \times \mathcal{C}$
 - $Y \in \mathbb{R}$: outcome
 - $\mathbf{X} \in \mathbb{R}^p$: continuous exposures
 - $\mathbf{C} \in \mathbb{R}^q$: covariates
- Data: i.i.d. sample $(\mathbf{O}_1, \dots, \mathbf{O}_n)$
- Potential outcomes: $Y^{\mathbf{x}} = \text{outcome under exposure } \mathbf{x}$
- Causal mean with intervention $\mathbf{x}$: $\mu_{\mathbf{x}}(\mathbf{x}) := \mathbb{E}[Y^{\mathbf{x}}]$
- Outcome regression: $m(\mathbf{x}, \mathbf{c}) := \mathbb{E}(Y \mid \mathbf{X} = \mathbf{x}, \mathbf{C} = \mathbf{c})$
- Propensity score: $\pi(\mathbf{x} \mid \mathbf{c}) = \partial P(\mathbf{X} \leq \mathbf{x} \mid \mathbf{C} = \mathbf{c}) / \partial \mathbf{x}$

Causal Identification Assumptions

1. **Consistency:** If $\mathbf{X} = \mathbf{x}$, then the observed outcome equals the potential outcome: $Y = Y^{\mathbf{x}}$
2. **Positivity:** Every exposure level has nonzero probability (given covariates): $\pi(\mathbf{x} \mid \mathbf{c}) \geq \varepsilon > 0$
3. **Conditional ignorability:** Given covariates, exposure is independent of potential outcomes: $Y^{\mathbf{x}} \perp\!\!\!\perp \mathbf{X} \mid \mathbf{C}$

Regression adjustment / g -formula

$$\mu_{\mathbf{x}}(\mathbf{x}) = \mathbb{E}[m(\mathbf{x}, \mathbf{C})]$$

Identification of the Causal Mean

Regression adjustment / g -formula

$$\mu_{\mathbf{x}}(\mathbf{x}) = \mathbb{E}[m(\mathbf{x}, \mathbf{C})]$$

Generalized Propensity Score (GPS)

$$\mu_{\mathbf{x}}(\mathbf{x}) = \lim_{h \rightarrow 0} \mathbb{E} \left[\frac{Y \cdot K\left(\frac{\mathbf{X} - \mathbf{x}}{h}\right)}{h^p \cdot \pi(\mathbf{X} \mid \mathbf{C})} \right]$$

We focus on the g -formula since high-dimensional multivariate response conditional densities are almost impossible to estimate well.

Sufficient dimension reduction

Sufficient Dimension Reduction (SDR)

Goal: find a linear transformation $\beta \in \mathbb{R}^{p \times d}$ that retains all information about Y in $\mathbf{X}$

Central Subspace (CS):

$$P(Y \leq y \mid \mathbf{X} = \mathbf{x}) = P(Y \leq y \mid \beta^\top \mathbf{X} = \beta^\top \mathbf{x})$$

- Only $\text{span}(\beta)$ is identifiable
- $\mathcal{S}_{Y|\mathbf{X}} = \text{span}(\beta)$

Central Mean Subspace (CMS):

$$\mathbb{E}(Y \mid \mathbf{X} = \mathbf{x}) = \mathbb{E}(Y \mid \beta^\top \mathbf{X} = \beta^\top \mathbf{x})$$

- Only $\text{span}(\beta)$ is identifiable
- $\mathcal{S}_{\mathbb{E}(Y|\mathbf{X})} = \text{span}(\beta)$

Semiparametric Estimation of the CMS

- **Semiparametric regression model:**
 - Characterized by $\mathbb{E}(Y \mid \mathbf{X})$ depending only on $\beta^\top \mathbf{X}$
 - Parameter of interest: $\text{span}(\beta)$
- **Influence-function-based framework:**
 - Bickel et al. (1993), Tsiatis (2006) $\rightarrow$ full class of RAL estimators
 - Ma et al. (2014) derived efficient score for β
 - Luo et al. (2014, 2016) extended to other functionals and improved efficiency
- **Advantages:**
 - Doubly robust estimating equations
 - Asymptotic linearity for inference
 - Efficiency guarantees via efficient score

Gradient-based Alternatives

- **Key identity from CMS:**

$$\frac{\partial \mathbb{E}(Y \mid \mathbf{X} = \mathbf{x})}{\partial \mathbf{x}} = \beta \frac{\partial \mathbb{E}(Y \mid \beta^\top \mathbf{X} = \beta^\top \mathbf{x})}{\partial (\beta^\top \mathbf{x})} \in \text{span}(\beta)$$

- **Approach:** estimate gradients instead of full nuisance functions
- **Examples:**
 - Principal Hessian Directions (Li, 1992)
 - Minimum Average Variance Estimator (MAVE, Xia 2002)
- **Dimension selection:**
 - Simple cross-validation for d_0
 - Balances model fit and parsimony

Estimation with MAVE

Model:

$$Y = g(\beta^\top \mathbf{X}) + \varepsilon, \quad \beta \in \mathbb{R}^{p \times d}, \quad d \ll p$$

Local linear approximation:

$$Y_i \approx a_j + \mathbf{b}_j^\top \beta^\top (\mathbf{X}_i - \mathbf{X}_j)$$

MAVE objective:

$$\min_{\beta^\top \beta = I_d, \{a_j, \mathbf{b}_j\}} \sum_{j=1}^n \sum_{i=1}^n \left(Y_i - \{a_j + \mathbf{b}_j^\top \beta^\top (\mathbf{X}_i - \mathbf{X}_j)\} \right)^2$$
$$K_h(\beta^\top \mathbf{X}_i - \beta^\top \mathbf{X}_j)$$

Causal SDR: Motivation

- Standard SDR methods recover subspaces of $\mathbb{E}(Y \mid \mathbf{X})$
- With **confounding**, $\mathbb{E}(Y \mid \mathbf{X}) \neq \mathbb{E}(Y^{\mathbf{x}})$
- Goal: find $\beta \in \mathbb{R}^{p \times d}$ such that

$$\mu_{\mathbf{x}}(\mathbf{x}) = \mu(\beta^{\top} \mathbf{x}),$$

where $\mu_{\mathbf{x}}(\mathbf{x}) = \mathbb{E}(Y^{\mathbf{x}})$

- The estimand of interest is the **span of** β , called the *causal central mean subspace*

Semiparametric Estimation in the Causal Setting

- Nabi et al. (2022) extended semiparametric SDR to handle confounding
- Approximate efficient score:

$$S_{\text{eff}}(\beta) \approx \mathbb{E}_q[\mathbb{E}[\tilde{U}(\beta) \mid \mathbf{X}, \mathbf{C}] \mid \mathbf{C}] + \frac{\pi^*(\mathbf{X})}{\pi(\mathbf{X} \mid \mathbf{C})} \left(\tilde{U}(\beta) - \mathbb{E}[\tilde{U}(\beta) \mid \mathbf{X}, \mathbf{C}] \right)$$

Challenge: Requires estimation of many nuisance functions: regression, variance, conditional moments, and the generalized propensity score

- **Idea:** adapt MAVE (Minimum Average Variance Estimation) to causal SDR
- Note: $\mu_{\mathbf{x}}(\mathbf{x}) = g(\beta^\top \mathbf{x})$ for some smooth g
- Replace Y in the MAVE objective with estimated causal mean $\hat{\mu}_{\mathbf{x}}(\mathbf{X})$
- $\hat{\mu}_{\mathbf{x}}$ estimated once via the g -formula using flexible learners
- Then run MAVE optimization with $\hat{\mu}_{\mathbf{x}}$ treated as fixed
- Benefits:
 - Consistent for the causal central mean subspace
 - Efficient iterative algorithm with closed-form updates. No nuisance estimation or high-dimensional search.

- **Convergence:**

$$\|(I - \hat{\beta}\hat{\beta}^\top)\beta\| = O_p(h^3 + h\delta_n + h^{-1}\delta_n^2 + h^{-1}B_n)$$

where $\delta_n = (\log n / (nh^p))^{1/2}$, and B_n is the rate for $\hat{\mu}_{\mathbf{x}}$. If the causal mean is estimated well, csPCA converges faster than the normal kernel bias of $O(h^2)$.

- **Dimension selection procedure:**

- For each candidate d , compute $\hat{\beta}_d$ from csPCA
- Fit a kernel regression of $\hat{\mu}_{\mathbf{x}}(\mathbf{X}_i)$ on $\hat{\beta}_d^\top \mathbf{X}_i$
- Define CV score:
$$\text{CV}(d) = \frac{1}{n} \sum_i (\hat{\mu}_{\mathbf{x}}(\mathbf{X}_i) - \hat{a}_{d0,i})^2$$
- Select $\hat{d} = \arg \min_d \text{CV}(d)$ (consistent for the true dimension)

Simulation studies

Simulation DGP

Below is the data-generating process where $\Phi(\cdot)$ is the normal CDF and $\Sigma_X[i, j] = 0.8^{|i-j|}$.

$$\mathbf{C} \sim \mathcal{N}_q(\mathbf{0}, \Sigma_C)$$

$$\mathbf{X} \mid \mathbf{C} \sim N_p(\Phi(\Theta^\top \mathbf{C}), \Sigma_X)$$

$$Y \mid \mathbf{X}, \mathbf{C} = 0.1 \cdot [h(\mathbf{Z}) + g(\mathbf{C}) + Z_1 C_1] + \varepsilon_Y \sqrt{0.5 + \Phi(C_1 + C_2)}$$

The true causal mean is

$$\mu_\beta(\mathbf{z}) = 0.1 \cdot h(\mathbf{z}) = 0.1[z_1 + 0.5 \cdot (z_2^2 + z_1 z_2)]$$

Estimation error in $\mathcal{S}_{\mathbb{E}(Y^{\mathbf{x}})}$

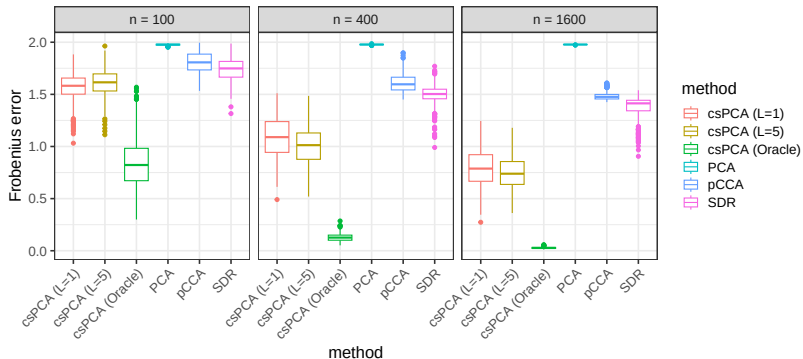


Figure 1: Frobenius error in estimating the causal central mean subspace.

Order determination

Table 1: Percentage of estimated structural dimension. In this simulation the true dimension was $d_0 = 2$.

n	Method	$d=1$	$d=2$	$d=3$	$d=4$
100	SDR	0.016	0.550	0.397	0.036
	csPCA (Oracle)	0.498	0.502	0.000	0.000
	csPCA (L=1)	0.343	0.605	0.052	0.000
	csPCA (L=5)	0.163	0.738	0.099	0.000
1600	SDR	0.784	0.214	0.002	0.000
	csPCA (Oracle)	0.000	1.000	0.000	0.000
	csPCA (L=1)	0.000	1.000	0.000	0.000
	csPCA (L=5)	0.000	1.000	0.000	0.000

Real data analysis

Dataset: Maternal and Fetal Cytokines

- **Study population:**
 - $N = 106$ women receiving elective pregnancy termination
 - 15–24 gestational weeks (2010–2016, UCSF clinic)
- **Data collected:**
 - Maternal sociodemographic and medical information
 - Biospecimens: maternal blood, matched cord blood samples
- **Measurements:**
 - 20 highly correlated pro- and anti-inflammatory cytokines
 - Outcome of interest: **cord blood IL-6**
- **Motivation:**
 - Mid-pregnancy fetal samples are rare
 - Opportunity to study maternal biomarkers → fetal inflammation

Correlation among exposures and outcome

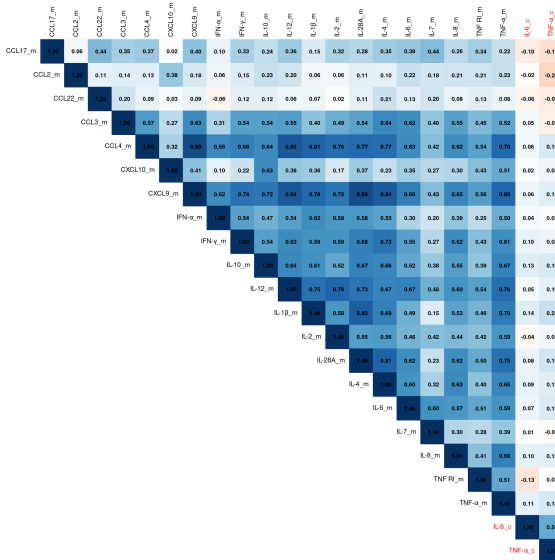


Figure 2: Correlation among maternal exposures and cord blood

Loadings (csPCA)

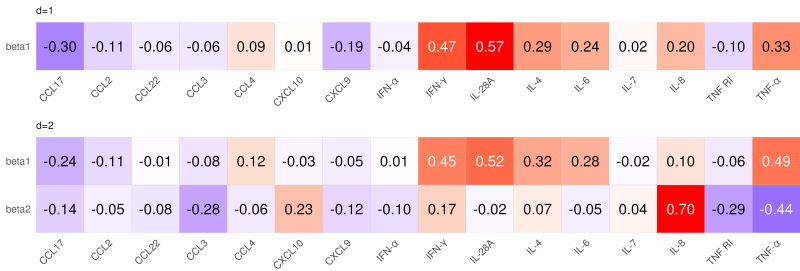


Figure 3: Estimate of β for $d = 1$ and $d = 2$. Can interpret as loadings in factor analysis or probabilistic PCA.

SDR relationship with causal mean

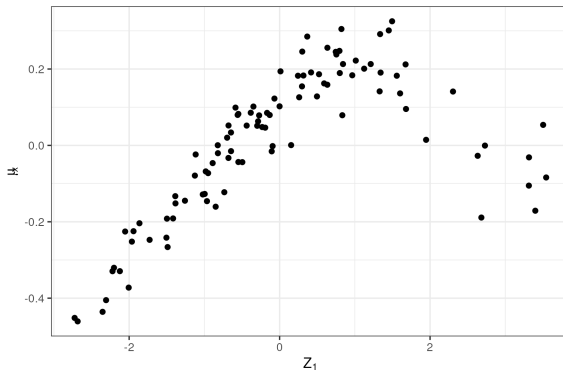


Figure 4: Scatterplot of the dimension reduction $Z = \beta^\top \mathbf{X}$ against the estimated causal mean $\hat{\mu}_{\mathbf{x}}(\mathbf{X}_i)$

- We propose a more computationally efficient method to estimate the causal central mean subspace
- This dimension reduction can improve on mixtures analysis by being both **causal** and **interpretable**
- More work! In particular efficiency gains, other dimension reduction frameworks, and theoretical guarantees

Acknowledgments

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Questions?

Lieshout, M.n.m. van. 2021. “Infill Asymptotics for Adaptive Kernel Estimators of Spatial Intensity.” *Australian & New Zealand Journal of Statistics* 63 (1): 159–81.
<https://doi.org/10.1111/anzs.12319>.

Appendix

Loadings (regular SDR)

(Lieshout 2021)

